Source-Grounded Multimodal Evidence Review for Public Defenders

Video Scan Pipeline Achieves 6.7× F1 Improvement over Transcript-Only Baseline

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NVIDIA Nemotron Omni

Evaluated on L40S GPU

github.com/harishvs/counterpoint

1. Problem & Motivation

Public defenders in the U.S. handle **80%** of criminal cases but are chronically under-resourced. A single attorney may manage hundreds of cases, each with hours of bodycam footage and multi-page incident reports.

The Gap

- Officers' reports may contain discrepancies with video evidence
- Manual cross-referencing takes hours per case
- Missed discrepancies can affect case outcomes

Our Goal

An AI tool that **automatically flags potential conflicts** between incident reports and bodycam footage, with source references for attorney verification.

Non-goals: determining guilt, assessing misconduct, deciding credibility, or replacing attorney review.

2. Benchmark Dataset

71	1,010	18
Scenarios	Claims	Source Videos

Each scenario pairs a **real bodycam video** with a fictional incident report containing planted discrepancies.

Ground Truth Distribution

Scenario Types

- 15 OCDA officer-involved shooting investigations
- 3 American Fork PD (trespass, harassment, stalking)
- 10 negative examples (fully accurate reports)
- 5 adversarial (near-correct with subtle errors)
- Multiple reports per video with different discrepancy profiles

Sources: OCDA & Internet Archive. Evidence not redistributed; manifest provides public download URLs with SHA-256 hashes.

3. Pipeline Evolution

We iterated through three architectures. Each solved a specific bottleneck, yielding a **6.7× improvement in F1** from baseline to final pipeline.

Architecture A: Transcript-Only (Baseline)

Report → Extract Claims → Match to Transcript → Verify (text only)

Bottleneck: No visual evidence. Model can only assess claims about speech, not positions, objects, or actions. F1 = 5.3%

Architecture B: Claim-First with Video Clips

Report → Extract → Keyword Retrieve → Cut Clip (ffmpeg) → Verify + Video

Bottleneck: Only ~30% of claims matched to transcript timestamps. Visual claims (positions, objects) have no audio keyword to match. F1 = 12.7%

Architecture C: Video Scan (Final)

Report → Extract → Chunk Video (2-min segments) → Scan ALL claims per chunk

Fix: Every claim evaluated against every video chunk. Visual claims now have evidence. Zero abstention. F1 = 35.4%

F1 Progression

4. Deployment Architecture

Designed for office-controlled AI: DGX Spark is our preferred on-premises target for reviewing sensitive discovery without sending case content to external AI services.

	Experimental	Proposed
Hardware	L40S (48 GB)	DGX Spark (128 GB)
Location	Cloud GPU	Defender office
Status	**Evaluated**	Proposed target

Component	Technology
Model	Nemotron Omni 30B-A3B (NVFP4, 21 GB)
Serving	vLLM 0.20 + video/audio
Input	Video + audio + text, 2 min/chunk
License	NVIDIA Open Model (commercial OK)

Local deployment supports office control over sensitive evidence; confidentiality and compliance also depend on system configuration, access controls, and approved operating procedures.

5. Results (1,010 Claims)

Pipeline Comparison

Pipeline	Model	P	R	F1
Transcript-only	Claude Haiku	40.0%	2.8%	5.3%
Claim-first clips	Nemotron	62.5%	7.1%	12.7%
Video scan	**Nemotron**	**61.0%**	**24.9%**	**35.4%**

6.7×	0%	100%
F1 vs baseline	Abstention	P on 6 of 8 types

By Difficulty

Difficulty	P	R	F1	n
Easy	38.3%	23.6%	29.2%	532
Medium	94.5%	24.3%	38.7%	298
Hard	66.7%	28.8%	40.2%	180

By Discrepancy Type (ranked by F1)

Type	P	R	F1	n
Sequence mismatch	**100%**	**35.5%**	**52.4%**	**31**
Minimization	**100%**	**34.7%**	**51.5%**	**72**
Omission	100%	28.3%	44.1%	53
Fabrication	93.3%	26.4%	41.2%	54
Temporal mismatch	100%	23.4%	37.9%	78
Exaggeration	100%	21.8%	35.8%	78
Direct contradiction	100%	20.9%	34.5%	208
Ambiguous language	16.7%	33.3%	22.2%	76

6. Confusion Matrix

Video scan, Nemotron Omni, 1,010 claims

	ambiguous	missing	conflict	supported
Conflict (590)	**289**	58	**147**	96
Supported (347)	186	23	**77**	**61**
Ambiguous (67)	35	9	15	8
No evidence (6)	3	0	2	1

147 true conflicts detected. 77 false positives from supported claims. 289 conflicts missed as ambiguous—the primary area for improvement.

7. Key Findings

- **Video scanning solved the retrieval bottleneck**: F1 jumped from 12.7% to 35.4% by scanning full video instead of keyword-matched clips
- **100% precision on 6 of 8 discrepancy types** when flagged: the model rarely makes false accusations
- **Hard claims outperform easy** (F1 40.2% vs 29.2%): medium/hard discrepancies have clearer visual signals
- **Nemotron's 2-min video limit** requires chunked architecture; validates against single-pass approaches
- **Zero abstention**: model commits on every claim with video evidence
- **DGX Spark proposed**: NVFP4 (21 GB) fits within 128 GB unified memory for on-premises inference

8. Limitations & Next Steps

Limitations

- 24.9% recall: 289 of 590 conflicts classified as ambiguous
- 77 false positives from supported claims (hurts precision for easy claims)
- Benchmark uses authored reports, not real police reports
- Ground truth derived from DA letters, not independent video review

Next Steps

- Prompt tuning to reduce false positives and ambiguous classifications
- Larger models (FP8/BF16) for improved reasoning
- Validate on DGX Spark and benchmark against L40S
- Seek defender office or legal-clinic collaborator

References

NVIDIA Nemotron Omni — build.nvidia.com
vLLM — github.com/vllm-project/vllm
OCDA OIS Reports — ocdistrictattorney.gov
Counterpoint — github.com/harishvs/counterpoint